

BRAC UNIVERSITY
Department of Computer Science and Engineering

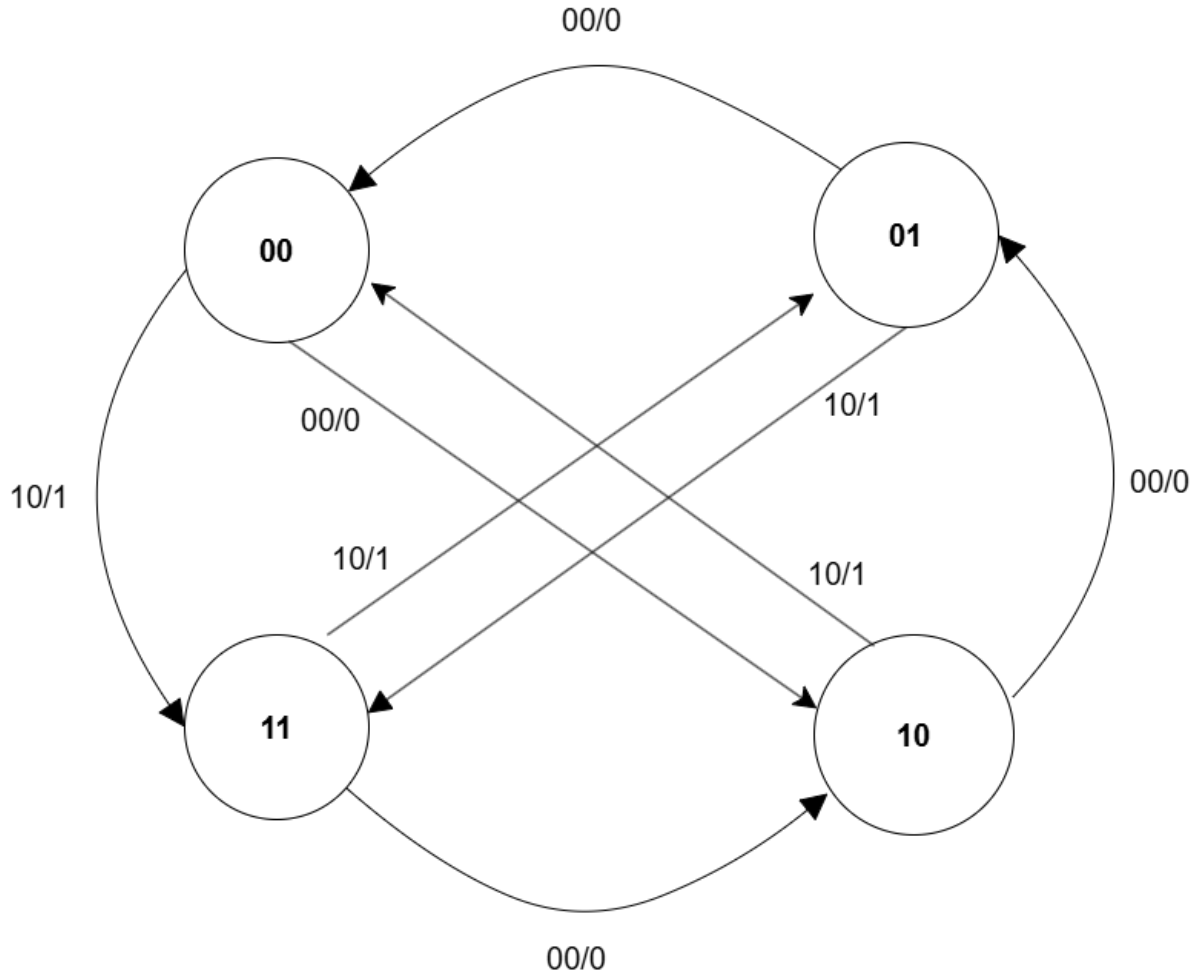
Examination: Makeup Quiz
Duration: 20 minutes

Semester: Summer 2025
Full Marks: 15

CSE 260: Digital Logic Design

Name:	ID:	Section:
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Given the state diagram as follows, get the sequential circuit using JK flipflop. [15 Mark]



For the missing input transitions, the given rules are to be followed:

- If the values of both external inputs are identical (i.e., 00 or 11), then the next state will be the same as the present state.
- If the values of the two external inputs are different (i.e., 01 or 10), then the next state will be determined by incrementing one to the present state. However, if the present state is the highest numbered state, the next state will transition to the initial state.

For determining missing outputs, the given rules are to be followed:

- If a state transition occurs from an odd-numbered state to another odd-numbered state, or from an even-numbered state to another even-numbered state, the external output will be 1.
- If a state transition occurs from an odd-numbered state to an even-numbered state, or from an even-numbered state to an odd-numbered state, the external output will be 0.